A Controller Comparison

Unit 4 · Topic 4.8 · TODAY'S PROBLEM

Suppose a food cooperative has 1,200 refrigerated display cases with conventional controllers and 900 with adaptive controllers that adjust cooling automatically. It takes independent random samples of 40 cases from each group; controller type was not assigned. A 95% interval for conventional minus adaptive mean daily use is $(1.219, 4.781)$ kWh per case. Replacing controllers is worthwhile only if mean savings are at least 2.0 kWh.

Rhea: "Zero is excluded and 2 is inside, so replacing controllers is shown to cause savings of at least 2 kWh."

Sol: "I would compare the interval with both 0 and 2, then check how the groups were formed."

Daily kWh samples

Controller	N	n	$\bar{x}$	s
Conv.	1,200	40	18.6	4.0
Adaptive	900	40	15.6	4.0

FIRST TAKE (5 min, on your sheet)

Does this range seem to promise savings of at least 2 kWh? Explain in one sentence.

DOK 1 **(a)** Locate 0 and 2 relative to the interval.

DOK 2 **(b)** Interpret the supplied 95% interval in one sentence about the two populations of display cases.

DOK 3 **(c)** In 3–4 sentences, decide whose account is justified. Use 0 and the 2-kWh cutoff, then explain why the way controllers were chosen limits a cause-and-effect claim.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.